

Dynamic allocation of window size detection of the low-rate DDoS attack

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Abstract:

Distributed denial of service (DDoS) attacks are a major concern in the world of cybersecurity. High-intensity and low-intensity DDoS attacks are the two most common varieties. Slowly but gradually, low-rate DDoS attacks have become the norm and significantly reduce network capacity. Low-rate DDoS attacks were previously detected using a predetermined window of time. The one drawback of the approach is that it is limited to a certain time frame. If no suspicious data packets are identified within that time, they will change the period statically. However, we were unable to identify complex attacks using this method since the attacker sends the attack packet at a predefined time. Recent studies have focused on identifying slow DDOS attacks. The window size is continually adjusted based on incoming data analysis, allowing for the detection of changing traffic patterns as the attack continues. To address this issue, we examined the performance of several machine learning and deep learning classifiers for dynamically allocating window sizes for DDoS detection. We test the proposed system on a publicly accessible network traffic dataset, proving its capacity to detect DDoS attacks while reducing false positives. Using the UNSW NB15 datasets, we evaluated different methods and shared our results. This study proposes a strategy for enhancing the accuracy and reliability of DDoS attack detection in practice by adjusting the window size at which network traffic characteristics are obtained.

Introduction:

The Internet's explosive expansion has some beneficial impacts, such as promoting economic development and cross-cultural communication, but it has also created new and significant threats to network security. A DDoS attack is one of the most dangerous types of cyberattacks that may occur today. Dozens, or even millions, of computers, may launch coordinated attacks on unsuspecting clients in what are known as distributed denial of service (DDoS) attacks. Detecting and mitigating DDoS attacks is a difficult problem for businesses, governments, and organizations of all kinds. Attacks using DDoS may be categorized as either high-intensity or low-intensity. These attacks aim primarily to slow down the rate at which packets enter the channel. The application-level Slow-Loris attack is not volume-dependent, unlike the high-intensity Smurf attack that includes the quick transfer of packets and results in a significant increase in traffic volume. Entropy computing is one approach to identifying DDoS attacks by measuring the unpredictability of incoming network data. Multi-classifier systems using entropy-based characteristics are another possibility for improved DDoS attack detection.

The phrase "window size" relates to how long network data is examined to detect DDoS attacks. A preset window size has traditionally been used in DDoS attack detection, which implies that a determined period is established, and network traffic data is evaluated during that interval to uncover suspicious patterns or features associated with attacks that use DDoS. The main problem with using a fixed window size is that it may miss some types of DDoS attacks, notably low-rate DDoS attacks. Low-rate DDoS attacks are defined by their slow and complex nature, making differentiation from authorized traffic within a certain time frame challenging. As a result, using a set window size may result in misleading results, such as false positives (mistaking legitimate traffic for an attack) and false negatives (failing to identify an actual attack).

To avoid this issue, we recommend utilizing a "dynamic window size." Instead of having a fixed period, the window size is dynamically adjusted based on data analysis and evolving traffic trends. This dynamic window size allocation allows the DDoS detection system to adapt to the varying rate at which data enters the network, making it more capable of effectively identifying low-frequency and complicated DDoS attacks. Even if the attack packets are broadcast at irregular intervals, the system can identify minor variations in traffic behavior caused by DDoS attacks by dynamically changing the window size in real-time. This strategy improves the accuracy and reliability of DDoS attack detection while reducing false positives and false negatives.

The main contribution of this research is, that we provide a novel approach to enhancing DDoS attack detection by using a dynamically allocated window size rather than a fixed one. This configurable window size, which is determined by examining incoming data, allows for better identification of changing traffic patterns during DDoS attacks. Secondly, compare several machine learning classifiers and deep learning approaches to find the most successful models for dynamic window size allocation. This evaluation aids in the detection of low-rate DDoS attacks while avoiding false positives and negatives.

Literature Review

For the recognition of the Low Rate of DDoS attacks, many methods have been introduced for this purpose, the authors from [1] propose utilizing machine learning classifiers to counteract DDoS attacks by developing E3ML, a feature-generating language for entropy-based features. Their method regularly achieves high accuracy in detecting various DDoS attacks by combining entropy-based features with machine learning techniques, delivering promising results. They do, however, compute the entropy of each traffic characteristic over a certain time, W , which in all trials is set to 60 seconds. The author chose a static timeframe is 60 seconds. because it was demonstrated to be an appropriate classification point [1]. Also, another paper [2] offers a hybrid deep neural network for LDoS detection that combines 1D convolutional neural networks with gated recurrent units. Real-world testing outperforms traditional signal analysis. The evaluation includes running attack software 100 times and recording real-world LDoS attacks in a lab setting. Also, another paper [3] provides a real-time traffic analysis technique that splits network trace data into time intervals using thresholds. It takes log data and converts it into feature vectors containing the average hop count, payload, access count, and latency. However, depending just on traffic data to detect L-DoS attacks may be ineffective. Also, another paper [4] describes a clustering-based approach for identifying network traffic that is prone to LDoS attacks. They employ UTR analysis to calculate the UDP/TCP traffic ratio inside clusters. The WIDE dataset, NS2 simulation platform, and network test-bed environment are utilized to assess the efficiency of the suggested detection method.

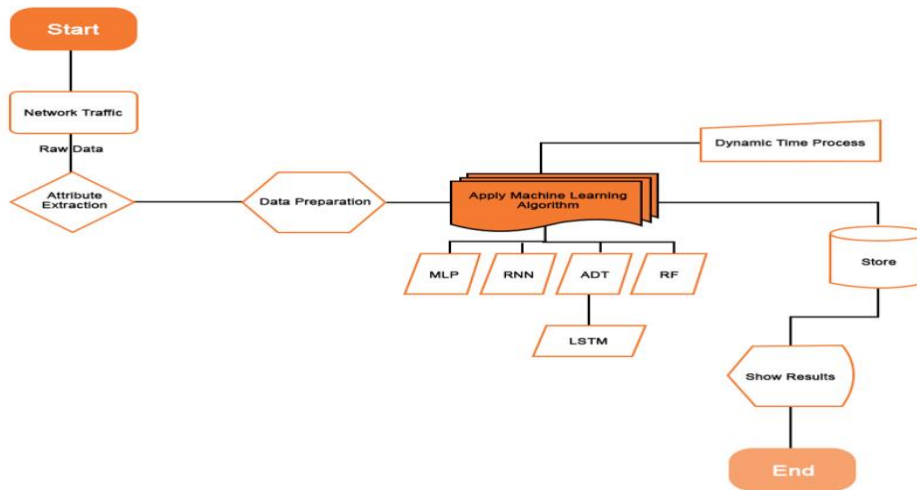


Fig. 1. System architecture – Attack detection Mechanism

Proposed Scheme

Attack detection using different machine learning algorithms (i.e Recurrent Neural Network (RNN), Multilayer Perceptron (MLP), Alternating Decision Tree (ADT) also using the different advanced variety of the machine learning algorithm i.e Long Short-Term Memory Networks (LSTM), Random Forest (RF)) using the voting system between different algorithm. The two sets of algorithms are a set of standard algorithms (MLP, RNN, RF, and LSTM) and a set of advanced algorithms. first, samples of both attack traffic and routine traffic are compiled into a dataset. This dataset is then divided into training and testing sets to ensure a fair evaluation of the algorithms. Then, using the training set, the standard algorithms (MLP, RNN, RF, and LSTM) are trained. The testing set is used to assess their performance following training. Evaluation metrics including accuracy, precision, recall, and F1 are used to assess each algorithm's effectiveness in detecting low-rate attacks. If the results demonstrate that the traditional algorithms provide results that are equal to or barely different from one another, it may be concluded from the analysis of the findings that their performance in recognizing low-rate assaults is comparable. In these situations, it is crucial to use a third technique called ADT (Alternate Decision Tree) to more thoroughly evaluate the attack traffic. The results of the ADT algorithm can be compared to those of the common methods to conclude the existence of attack traffic.

Expected outcomes

Examined the performance of the solution in experimental assessment by altering the number of rows in the data sample that we utilized to detect attacks. Looked at three alternative scenarios: 100 rows, 1000 rows, and 10,000 rows. After cleaning and processing the data, we go on to exploratory data analysis, where we visualize and summarize the data to get insights into the relationships between different aspects. This requires creating histograms or density plots to display numerical feature distributions and comparing these distributions across several categories using boxplots. Furthermore, employ bar charts to depict the frequency of various categories within category characteristics. Summary statistics such as mean, median, and standard deviation are created for numerical attributes.

Conclusion

The authors deliver a unique approach for detecting DDoS attacks that dynamically adapts the detection window size, hence boosting flexibility to changing traffic patterns. Extensive examination of public network traffic data using different machine learning and deep learning classifiers yields promising results for identifying DDoS attacks while minimizing false positives. Continuous monitoring and window size change efficiently capture changing attack patterns, potentially improving network security and minimizing DDoS attacks on vital systems and services.

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